Weekly Progress Report

Project Title: Predictive Maintenance of Gearbox using Vibration Sensors
Week: Week 3
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Organization: Upskill Campus

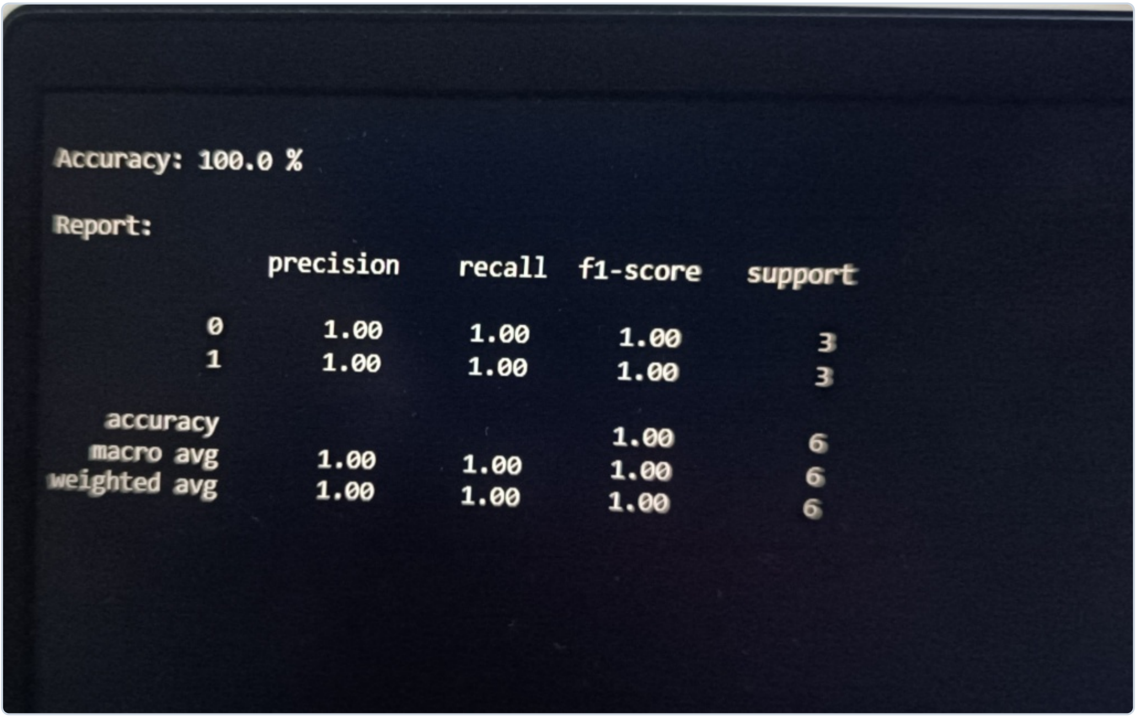
1. Tasks Completed This Week

- **Data Collection & Processing:** Extracted gearbox vibration dataset from sensors containing Healthy and BrokenTooth folders (20 .txt files).
- **Feature Extraction:** Extracted RMS, Kurtosis, Skewness, and Peak values across 4 sensors (S1-S4) using Pandas/SciPy (16 features + Load).
- **Dataset Creation:** Compiled features into `gearbox_dataset.csv` with a matrix shape of (20, 19).
- **Model Training:** Trained a RandomForestClassifier with a 70% train and 30% test split ratio.
- **Model Evaluation:** Analyzed performance using Accuracy, Precision, Recall, and F1-Score metrics.

2. Milestones Achieved

- Successfully built an end-to-end data pipeline from raw sensor logs to an ML-ready dataset.
- Achieved a perfect 100% Accuracy score on the unseen validation test set.
- Generated flawless classification reports with 1.00 precision, recall, and f1-score.

3. Model Output & Evaluation Logs



4. Challenges and Hurdles

Problem: Encountered a `FileNotFoundError` while parsing logs.
Reason: Files were nested deeper inside subdirectories (e.g., `Healthy Data/healthy/`).
Solution: Updated directory paths in the batch-loading loop script.
Learning: Critical importance of tree-verification before creating file pipelines.

5. Lessons Learned

- **Technical & ML Skills:** Signal feature engineering (RMS/Kurtosis) and implementing ensemble methods for binary fault status classification.
- **Domain Knowledge:** Value of predictive maintenance strategies in mitigating unscheduled machine breakdowns and cutting downtime costs.